Website layers: JS(behavior) CSS(presentation) HTML(content or structure)
Version control: github(most established platform) benefit(track code history, allow human error)
Github: repository, clone, remote, pull, push, fork, fetch, pull request, branch
Fork—clone—update—commit—push—send pull request; fetch—merge—push

Usability:
specific users – achieve goals with effectiveness, efficiency, satisfaction – specified context of use
users – interaction; context of use – phy/soc environment used by product; goal: outcome
Effectiveness: accuracy and completeness. Quantify by binary(tick-box) or spectrum
Efficiency: save time.
Satisfaction: user impression

Usability model: Nielsen, ISO, PACMAD
Traditional vs wide range vs mobile
其他两个比 ISO 多了 errors, memorability, learnability, + cognitive load
Learnability: easy to learn, get work rapidly
Memorability: easy to remember. 很久没用但很快想起来怎么用；用户不想删除
Cognitive load: amount of cognitive process required. 同时做很多事

User case: iterative strategy. Sequence of actions perform by system to yield observable result of value to a particular user.

UI: phy control; what look like; how input; how respond; how output.
Layout: network; mobile app; navigation control; split view; tab view; page view
Button; Switch; Slider (predefined scale)
Dialogue window (info+ask action)
Keyboard on TV(alphabetic layout – reduce typing)
Process bar (give prompt response)
Media <iframe width = x , height = y, src = “www…”> </iframe>

Wireframe: rough sketch; mock-up: add details for visual effects; prototype: close to finished product, can be simulated and tested.

Design principle: simplicity, structure, consistency, tolerance
Simplicity: minimalism. (open space, level of extremism, reduce excessive express, depart from nature); not opposite to complexity
Structure: organization of UI meaningfully. (Navigation; use of control) flat: fewer main, extended to children; deep: more main, fewer to children. Breadcrumb (trace: wherever + get back)
Consistency: in appearance, placement, and behavior
Tolerance: prevention: ask confirm 提前写的提示; recoverability: forward: accept and help; backward: undo

Mobile compatibility: objects presented as desired on all displays
Responsive web design:

<meta name=”viewport” content=”width=device-width, initial-scale=1.0”> head section
Width = 100%
@media only screen and (max-width:620px)
Touch vs Mouse build-in, gesture, available, intuitive vs accurate, hover, change shape

Mobile device input: sensor; touch screen; buttons; camera; GPS. Output: speaker, vibrator
Advan: portability, sensors, portrait orientation, gestures, niche
Limit: context, slow connect, size, resolution, power, entry method

Fragmentation: versions; devices
Quality control: approve before release. Remove. Still bad.

Interactivity: ability to influence communication (change input way, change context way)
Cognitive psychology: mental action of acquiring knowledge, understanding (how perceive, how process and store, how manipulate)
Perception: active, blend knowledge and sensation (depend on what know, what expect)

Gestalt: see objects by group, recognize, simplify
Proximity: group by distance
Similarity: by appearance
Continuity: by flow or alignment
Closure: perceive shape not complete
Ground and figure: isolate into background and fore
Symmetry: by meaning
Pop-out: spatial similar, but not part

Memory: sensory buffer by attention – working memory by rehearsal – long-term memory by recall
Sensory buffer: receive stimuli, 0.5s, store or overwrite+lost
Working memory: active state, transient, 5-9 chunks, lost without rehearsal
Chunks: constrain short-term memory
Long-term memory: meaning and knowledge. Meaningful info+prediction
Recognition: item is presented with what you try to remember.
Recall: remember something.
Easy to recall meaning than exact word
Minimize cognitive load (amount of working memory resources used to perform tasks): chunk, recognition, visual clue, memory aids, keep simple, avoid visual clutter, short-term memory support.
Emphasis: remind good parts, create habits, personalize

Sound active (cut in), graphics passive (must look around, can overlook)
Sound non-localized (attract), graphics localized (localize attention)
Sound transient (repetition), graphics permanent
Sound design: volume, length, lead-in, sets

Color perception (subjective, by context, by experience and function)

Color association (built-in, cultural, work)
Color usage: decorate, group, highlight, color code (blue links), measure
Color scheme (key, monochromatic, complementary and adjacent)
Mono: single hue, different brightness
Complementary: hue opposite on wheel. Striking and distinctive
Adjacent: subtle impression

Video: telling, show change, demonstrate process (size vs quality)
Video control: depend on context, length, message, tasks

Text properties: easy generate, split, analyze by computer, print out, download, give structure
Fonts: consistent
Text structure (short para, many head, minimize scroll)
Text consistency (style, knowledge level, address, term)

UX talents: emotions, motivations, cognitive load, memory, subconsciousness
Emotion: what they feel. Positive: gain. Negative: loss. Provide signal to feel useful emotion
Emotional design: trigger emotion which contribute connection between user and product (visceral, reflective, behavioral)
Motivation: activate, direct and sustain users' behavior (curiosity, affiliation, status, justice)
Curiosity: gain or loss so hold some back. E.g. apple website. Less is more.
Affiliation: belong to a group, be identified by common things
Status: control, compare, compete (control setting, show measurement)
Justice: have rules, respect symbols, right to choose champions
Memory: 在上面
Common principle for subconsciousness: cognitive bias (anchoring, bandwagon, decoy); illusion of choices (paradox choice, give option carefully, choose default, comparisons); hyperbolic discounting(choose smaller-sooner rewards, destructive design)

Gamification: encourage engagement: integrate reward and punish through goals – feedback loops, progressive challenges.

Interactive media design: integration of digital media + into a structured computerized environment + allow people to interact with data + for purposes
Design process:
Information phase: gather info:
1. determine goal of the object (意图)
2. gather data on target audience (谁？怎么获取 app？什么环境？)
3. select content for project (scope and depth) (brainstorm and mind map)
Interaction: how interact and navigate in app:
1. determine content flow (how to receive content) (group by object and work out a storyboard)
2. play controls (by users) (哪些控件可以用)
3. map navigation structure (overview of content of storyboard, reference of navigation flow)
Presentation: how app look:

1. create layout thumbnails (general layout by thumbnails) – small drawing, test arrangement, decide working best one
2. design the look of the project (style, how look) (由导航和布局、目的、用户所影响)
9. build prototype (surface issue. Follow up with end users and manage their expectation)

Accessibility: equal access
Equity: fairness and justice
Diversity: variation of different characteristics of people
Inclusive design: consider full range of human diversity. Why important: increase sales, rank higher in engine, boost a brand.
Principles: identify ability-based exclusion, identify situational challenges, avoid personal biases, offer ways to engage, provide equal experience to users
User personas: create based on research, represent different user types, understand users' needs, build empathy, from observations, map needs to functionality.
Why important: build empathy; gain perspective; identify with user.
Persona: define who the story is about.
Goal: what the persona wants to fulfill.
scenario: when where and how take place.
Good persona: tie to real data; reflect real pattern; focus on current; context specific.
Persona spectrums: take typical user – create iterations for same persona – factor wide range of user needs.
Consider permanent, temporary, situational scenarios.

Heuristic analysis principles:
H1 visibility of system status keep users informed
H2 match between system and the real world speak user's language
H3 user control and freedom allow mistaken choices, undo, redo
H4 consistency and standards within and between applications
H5 error prevention help undo; go back quickly; change their minds
H6 recognition rather than call make things visible and retrievable
H7 flexibility and efficiency of use accelerators, tailor frequent actions
H8 aesthetic and minimalist design delete useless design
H9 help users recognize, diagnose, and recover from errors
Clearly indicate; human-readable; polite; describe problems; give advice; be noticeable; preserve work; guess correct action; educate by linking
H10 help and documentation easy to search; focus on task; concrete steps

Meet expectations user focused errors keep it simple

User testing (usability testing, tree testing, remote usability testing, open card/closed card/remote card sorting, focus groups, qualitative attitudinal/behavioral user testing)
Limit: not tell how to fix problem; not understand reasons behind reactions; difficult to generalize analysis of entire system

Support statements with evidence:
Include:
feedback of survey, interview…
verdict from experts…
research outcome from existing publications